ETALE SURJECTIVE MORPHISMS OF LOCALLY NOETHERIAN FORMAL SPACES ARE RIG-SURJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof base-changes the test object, extracts an affine etale piece, finds a maximal ideal over the closed point, promotes its local ring to a complete DVR, and composes the resulting formal spectrum morphisms

Editorial note. The label lemma-etale-covering-rig-surjective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- rig-surjectivity tested by adic morphisms from complete discrete valuation rings
- base change of representable etale morphisms and nonemptiness of a surjective etale cover
- etale morphisms of formal algebraic spaces correspond to etale ring maps on formal spectra (Formal Spaces, Lemma formal-spaces-lemma-etale)
- etale local rings over a DVR are discrete valuation rings (More on Algebra, Lemma more-algebra-lemma-Dedekind-etale-extension) and their completions are complete DVRs (More on Algebra, Lemma more-algebra-lemma-completion-dvr)
- morphisms between formal spectra (Formal Spaces, Lemma formal-spaces-lemma-morphism-between-formal-spectra)

Source reference. Original lemma 0AN8: sources/stacks-chow/tags.txt, source lines 13431–13431. Prerequisite pointer invoked at proof line(s) 5377; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AN0: sources/stacks-chow/tags.txt, source lines 13423–13423. Prerequisite pointer invoked at proof line(s) 5385; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AP2: sources/stacks-chow/tags.txt, source lines 13460–13460. Prerequisite pointer invoked at proof line(s) 5381; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0AP1: sources/stacks-chow/tags.txt, source lines 13459–13459. Prerequisite pointer invoked at proof line(s) 5383; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of locally Noetherian formal algebraic spaces which is representable by algebraic spaces, étale, and surjective. Then f is rig-surjective.*

Proof. Let $p : \mathrm{Spf}(R) \to Y$ be an adic morphism where R is a complete discrete valuation ring. Let $Z = \mathrm{Spf}(R) \times_Y X$. Then $Z \to \mathrm{Spf}(R)$ is representable by algebraic spaces, étale, and surjective. Hence Z is nonempty. Pick a nonempty affine formal algebraic space V and an étale morphism $V \to Z$ (possible by our definitions). Then $V \to \mathrm{Spf}(R)$ corresponds to $R \to A^\wedge$ where $R \to A$ is an étale ring map, see Formal Spaces, Lemma 0AN8. Since $A^\wedge \neq 0$ (as $V \neq \emptyset$) we can find a maximal ideal $\mathfrak{m}$ of A lying over $\mathfrak{m}_R$. Then $A_\mathfrak{m}$ is a discrete valuation ring (More on Algebra, Lemma 0AP2). Then $R' = A^\wedge_\mathfrak{m}$ is a complete discrete valuation ring (More on Algebra, Lemma 0AP1). Applying Formal Spaces, Lemma 0AN0. we find the desired morphism $\mathrm{Spf}(R') \to V \to Z \to X$. $\square$